

Parrot

Parrots (order Psittaciformes), also known as **psittacines** (/ˈsɪtəsəmz/)^[1] from the name of the type genus *Psittacus*,^[2]^[3]^[4] are birds with a strong curved beak, upright stance, and clawed feet.^[a] They are classified in four families that contain roughly 410 species in 101 genera, found mostly in tropical and subtropical regions. The four families are the Psittaculidae (Old World parrots), Psittacidae (African and New World parrots), Cacatuidae (cockatoos), and Strigopidae (New Zealand parrots). One-third of all parrot species are threatened by extinction, with a higher aggregate extinction risk (IUCN Red List Index) than any other comparable bird group.^[5] Parrots have a generally pantropical distribution with several species inhabiting temperate regions as well. The greatest diversity of parrots is in South America^[6] and Australasia.^[7]

Parrots, along with corvids (ravens, crows, jays, and magpies), are among the most intelligent birds, and the ability of some species to imitate human speech enhances their popularity as pets. They form the most variably sized bird order in terms of length; many are vividly coloured and some, multi-coloured. Most parrots exhibit little or no sexual dimorphism in the visual spectrum.

The most important components of most parrots' diets are seeds, nuts, fruit, buds, and other plant material. A few species sometimes eat animals and carrion, while the lories and lorikeets are specialised for feeding on floral nectar and soft fruit. Almost all parrots nest in tree hollows (or nest boxes in captivity), and lay white eggs from which hatch altricial (helpless) young.

Trapping wild parrots for the pet trade, as well as hunting, habitat loss, and competition from invasive species, has diminished wild populations, with parrots being subjected to more exploitation than any other group of wild birds. As of 2021, about 50 million parrots (half of all parrots) live in captivity, with the vast majority of these living as pets in people's homes.^[8] Measures taken to conserve the habitats of some high-profile charismatic species have also protected many of the less charismatic species living in the same ecosystems.

Parrots are the only creatures that display true tripedalism, using their necks and beaks as limbs with propulsive forces equal to or greater than those forces generated by the forelimbs of primates when climbing vertical surfaces. They can travel with cyclical tripedal gaits when climbing.^[9]

Taxonomy

Origins and evolution

Psittaciform diversity in South America and Australasia suggests that the order may have evolved in Gondwana, centred in Australasia.^[10] The scarcity of parrots in the fossil record, however, presents difficulties in confirming the hypothesis. There is currently a higher number of fossil remains from the northern hemisphere in the early Cenozoic.^[11] Molecular studies suggest that parrots evolved approximately 59 million years ago (Mya) (range 66–51 Mya) in Gondwana. The Neotropical parrots are monophyletic, and the three major clades originated about 50 Mya (range 57–41 Mya).^[12]

A single 15 mm (0.6 in) fragment from a large lower bill (UCMP 143274), found in deposits from the Lance Creek Formation

Parrots

Temporal range: **Eocene–Holocene**



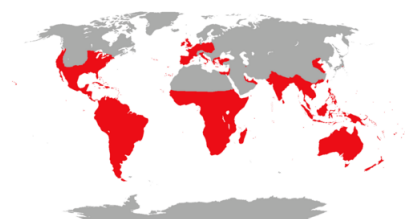
Rainbow lorikeet (*Trichoglossus moluccanus*)

Scientific classification ✎

Kingdom: Animalia
 Phylum: Chordata
 Class: Aves
 Clade: Psittacopasseres
 Order: Psittaciformes
 Wagler, 1830

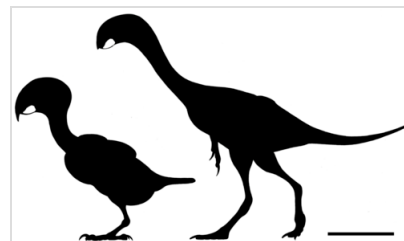
Superfamilies

Cacatuoidea (cockatoos)
Psittacoidea (true parrots)
Strigopoidea (New Zealand parrots)



Range of parrots, all species (red)

in Niobrara County, Wyoming, had been thought to be the oldest parrot fossil and is presumed to have originated from the Late Cretaceous period, which makes it about 70 million years old.^[13] However, other studies suggest that this fossil is not from a bird, but from a caenagnathid oviraptorosaur (a non-avian dinosaur with a birdlike beak), as several details of the fossil used to support its identity as a parrot are not actually exclusive to parrots, and it is dissimilar to the earliest-known unequivocal parrot fossils.^{[14][15]}



Fossil dentary specimen UCMP 143274 restored as a parrot (left) or an oviraptorosaur

It is generally assumed that the Psittaciformes were present during the Cretaceous–Paleogene extinction event (K-Pg extinction), 66 mya. They were probably generalised arboreal birds, and did not have the specialised crushing bills of modern species.^{[11][16]} Genomic analysis provides strong evidence that parrots are the sister group of passerines, forming the clade Psittacopasserae, which is in turn the sister group of the falcons.^[17]

The first uncontroversial parrot fossils date to tropical Eocene Europe around 50 mya. Initially, a neoavian named *Mopsitta tanta*, uncovered in Denmark's Early Eocene Fur Formation and dated to 54 mya, was assigned to the Psittaciformes. However, the rather nondescript bone is not unequivocally psittaciform, and it may rather belong to the ibis genus *Rhynchaeites*, whose fossil legs were found in the same deposits.^[18]

Several fairly complete skeletons of parrot-like birds have been found in England and Germany.^[19] These are probably not transitional fossils between ancestral and modern parrots, but rather lineages that evolved parallel to true parrots and cockatoos.^[20]

- *Psittacopes*
- *Serudaptus*
- **Halcyornithidae**
 - *Cyrilavis*
 - *Halcyornis*
 - *Pulchrapollia*
 - *Pseudasturides*
- **Vastanavidae**
 - *Vastanavis*
- **Quercypsittidae**
 - *Quercypsitta*
- **Messelasturidae**^[21]
 - *Messelastur*
 - *Tynskya*



Cyrilavis colburnorum, from the Eocene Green River Formation in Wyoming

The earliest records of modern parrots date to around 23–20 mya.^[22] The fossil record—mainly from Europe—consists of bones clearly recognisable as belonging to anatomically modern parrots.^[23] The Southern Hemisphere contains no known parrot-like remains earlier than the Early Miocene around 20 mya.^[22]

Etymology

The name 'Psittaciformes' comes from the ancient Greek for parrot, ψιττακός (*'Psittacus'*), whose origin is unclear. Ctesias (5th century BCE) recorded the name *Psittacus* after the Indian name for a bird, most likely a parakeet (now placed in the genus *Psittacula*). Pliny the Elder (23/24–79 CE) in his *Natural History* (book 10, chapter 58) noted that the Indians called the bird "siptaces"; however, no matching Indian name has been traced.^{[24][25]} "Popinjay" is an older term for parrots, first used in English in the 1500s.^[26]

Phylogeny

Molecular phylogenetic studies have shown that Psittaciformes form a monophyletic clade that is sister to the Passeriformes:^{[27][28]} The time calibrated phylogeny indicates that the Australaves diverged around 65 Ma (million years ago) and the